

LEVEL 2 DATA SCIENCE

# Late Food-Delivery Order Prediction

A practical 2-4 hour machine learning project with Excel input and Excel output

DURATION

2-4 hours

MODEL TYPE

Binary classification

DELIVERABLE

Scored Excel file

LEVEL

Data Scientist L2

## Project snapshot

This project simulates a real operations analytics problem for a food-delivery company. You will use historical Excel data to train a model that predicts whether a new order is likely to be delivered late, then return a new Excel file containing prediction outputs and recommended actions.

|                                       |                                   |                                     |                                        |
|---------------------------------------|-----------------------------------|-------------------------------------|----------------------------------------|
| <b>500</b><br>labeled training orders | <b>120</b><br>new orders to score | <b>8</b><br>required output columns | <b>0.75+</b><br>minimum ROC-AUC target |
|---------------------------------------|-----------------------------------|-------------------------------------|----------------------------------------|

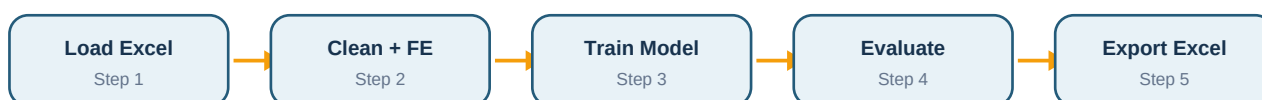
## Core challenge

Build a binary classification model that predicts **is\_late** before delivery completion. The model should generate a probability score, a late/not-late prediction, a risk band, and an operational recommendation for each new order.

Important leakage rule: do not use **actual\_delivery\_time\_min** as a feature. It is only known after the delivery is complete, so using it would make the model unrealistically strong and invalid.

## Project outcomes

- A trained classification model with meaningful validation metrics.
- A scored Excel output file for every order in the Scoring\_Orders sheet.
- Clear business interpretation using risk bands and recommended actions.
- A short explanation of the model, metrics, threshold, and top drivers of risk.



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| 2. Challenge definition       | Problem type, target variable, inputs, and leakage warning.   |
| 3. Workbook guide             | Explanation of each Excel sheet in the starter workbook.      |
| 4. Data dictionary            | Feature groups, usable fields, target, and forbidden columns. |
| 5. Recommended workflow       | Step-by-step implementation plan for 2-4 hours.               |
| 6. Modeling expectations      | Preprocessing, model options, thresholds, and validation.     |
| 7. Success metrics            | How to measure model performance and project quality.         |
| 8. Output Excel specification | Exact required columns and accepted values.                   |
| 9. Level 2 rubric             | What this measures for a Data Scientist Level 2.              |
| 10. Submission checklist      | Final review before returning the scored Excel file.          |

## Expected time split

| Activity                        | Suggested time | Output                                               |
|---------------------------------|----------------|------------------------------------------------------|
| Understand workbook + target    | 15-25 min      | Clear target and forbidden leakage columns.          |
| EDA + data cleaning             | 35-50 min      | Missing values handled and feature types understood. |
| Modeling pipeline               | 45-70 min      | Preprocessing + model trained on train split.        |
| Evaluation + threshold          | 30-45 min      | Metrics, confusion matrix, chosen threshold.         |
| Score new orders + export Excel | 30-45 min      | Final Excel file with 120 scored rows.               |

# 1. Business context

Food-delivery platforms lose customer trust when orders arrive late. Late orders can also increase refunds, complaints, support tickets, and driver reassignments. Operations teams need an early warning signal before the delivery is complete.

**Your model should act like a decision-support tool. It does not only predict; it also tells the business which orders deserve attention and what action should be taken.**

## Business question

For each new order, can we estimate the probability that it will be late before actual delivery completion?

## Potential business actions

- High-risk order: prioritize driver assignment, call the restaurant, or monitor live route status.
- Medium-risk order: monitor order and prepare escalation if the predicted risk increases.
- Low-risk order: no immediate action required.

## Why this is a good Level 2 project

It is small enough to complete in a few hours, but realistic enough to test important data science skills: data cleaning, leakage awareness, feature engineering, model validation, business thresholding, and Excel-based delivery of results.

### What this project measures



## 2. Challenge definition

| Item              | Definition                                        |
|-------------------|---------------------------------------------------|
| Problem type      | Binary classification                             |
| Target variable   | is_late                                           |
| Positive class    | 1 = order was late                                |
| Negative class    | 0 = order was not late                            |
| Unit of analysis  | One row = one food-delivery order                 |
| Prediction timing | Before actual delivery is completed               |
| Main output       | late_probability for each order in Scoring_Orders |

### Recommended definition of 'late'

Use the existing **is\_late** target in the training sheet. Do not recreate the target using post-delivery columns unless you are explicitly validating the data generation logic.

### What the model measures

- Operational delay risk based on order, restaurant, timing, weather, traffic, and driver-related variables.
- The probability that an order belongs to the late class.
- Whether the order needs intervention based on a chosen risk threshold.

### What the model does not measure

- It does not prove the exact reason why an order will be late.
- It does not replace human operations judgment.
- It should not use any information only available after delivery completion.

## 3. Excel workbook guide

Use the starter workbook **level2\_food\_delivery\_late\_prediction\_project.xlsx**. It contains the training data, new orders to score, output template, and guidance sheets.

| Sheet                | Purpose                 | How to use it                                                       |
|----------------------|-------------------------|---------------------------------------------------------------------|
| README_Project_Brief | Project instructions    | Read first. It explains the business problem and final deliverable. |
| Train_Orders         | Labeled historical data | Train and validate your model using this sheet.                     |
| Scoring_Orders       | Unlabeled new orders    | Score every row and produce model outputs.                          |
| Output_Template      | Required final format   | Match the final columns and naming exactly.                         |
| Data_Dictionary      | Column meanings         | Use it to identify feature types and leakage columns.               |
| Evaluation_Guide     | Metric targets          | Use it to assess whether your model is acceptable.                  |
| Project_Dashboard    | Starter summary         | Review high-level data profile before modeling.                     |

### Required file flow

| Input           | Process                                | Output           |
|-----------------|----------------------------------------|------------------|
| Train_Orders    | Clean, encode, split, train, evaluate  | Model + metrics  |
| Scoring_Orders  | Apply the same preprocessing and model | Scored orders    |
| Output_Template | Use the required column names          | Final Excel file |

### Deliverable naming suggestion

late\_delivery\_predictions\_<your\_name>.xlsx

## 4. Data dictionary and feature guidance

The exact workbook contains the full column-level dictionary. The table below explains the expected groups and how to think about them during modeling.

| Feature group         | Examples                                              | Use?        | Notes                                                      |
|-----------------------|-------------------------------------------------------|-------------|------------------------------------------------------------|
| Identifiers           | order_id                                              | No          | Keep for joining and export only. Do not train on raw IDs. |
| Order details         | order_value, item_count, distance_km                  | Yes         | Useful predictors of complexity and delivery time.         |
| Timing                | order_hour, day_of_week, is_weekend, peak_hour        | Yes         | Often strong predictors because demand changes by time.    |
| Restaurant operations | restaurant_rating, prep_time_est_min, restaurant_area | Yes         | Can capture preparation difficulty and local conditions.   |
| Driver features       | driver_experience_months, driver_rating               | Yes         | Use only if known before assignment or scoring time.       |
| External conditions   | weather_condition, traffic_level                      | Yes         | Important operational risk indicators.                     |
| Target                | is_late                                               | Target only | Available only in Train_Orders.                            |
| Leakage               | actual_delivery_time_min                              | No          | Known after delivery, so it invalidates the model.         |

### Leakage checklist

- Ask: would this value be known at the time I want to make the prediction?
- Remove direct outcomes such as actual delivery time, refund outcome, complaint outcome, or final delay minutes.
- Keep identifiers outside the feature matrix unless you deliberately create safe aggregate features from historical data.
- Apply preprocessing fitted on training data to scoring data; do not refit encoders separately on the scoring sheet.

A model with suspiciously perfect validation performance is usually a leakage problem, not a great model.

## 5. Recommended workflow

Follow this sequence to complete the project efficiently. The goal is not to test every model possible; the goal is to produce a reliable, explainable, business-ready Excel output.

- 1 Load the Excel workbook and inspect sheet names, row counts, and column types.
- 2 Separate Train\_Orders into X features and y target. Keep order\_id separately.
- 3 Remove leakage columns and target-related post-event variables from X.
- 4 Split Train\_Orders into train and validation sets using stratified sampling.
- 5 Build a preprocessing pipeline: impute missing numeric values, impute categorical values, and one-hot encode categoricals.
- 6 Train at least one baseline model, then optionally compare with a stronger tree-based model.
- 7 Evaluate ROC-AUC, recall for late orders, precision for predicted late orders, and confusion matrix.
- 8 Choose a classification threshold based on business need, not accuracy alone.
- 9 Score all rows in Scoring\_Orders and create the required output columns.
- 10 Export the final Excel file and verify row counts and column names.

### Suggested model sequence

| Model               | Why use it                                         | When it is enough                            |
|---------------------|----------------------------------------------------|----------------------------------------------|
| Logistic Regression | Simple baseline, interpretable, fast               | Good for checking pipeline correctness.      |
| Random Forest       | Handles nonlinear patterns and mixed features well | Good default for a Level 2 project.          |
| Gradient Boosting   | Often strong on tabular data                       | Use if you have time to compare performance. |



## 6. Modeling expectations

Your model should be reproducible and fair to the validation data. Use a pipeline so that the exact same cleaning and encoding logic is applied to both validation and scoring orders.

### Minimum modeling requirements

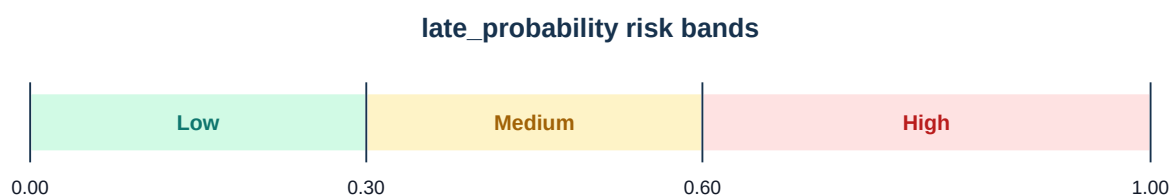
- Use a train-validation split, preferably 80/20 with stratification on `is_late`.
- Use a preprocessing pipeline rather than manual transformations that are hard to reproduce.
- Report metrics on the validation split only, not the training split only.
- Save or clearly define the threshold used to convert `late_probability` into `predicted_is_late`.
- Use the same selected feature columns for `Train_Orders` and `Scoring_Orders`.

### Suggested threshold logic

A default threshold of 0.50 is acceptable, but a business-focused threshold may be better. For late delivery, missing a truly late order can be costly, so you may lower the threshold to improve recall if precision remains reasonable.

| Threshold choice   | Effect                                                 | Use when                                           |
|--------------------|--------------------------------------------------------|----------------------------------------------------|
| Higher threshold   | Fewer late predictions, higher precision, lower recall | Operations team can handle only a few escalations. |
| Lower threshold    | More late predictions, higher recall, lower precision  | Business wants to catch more risky orders.         |
| Balanced threshold | Trade-off between precision and recall                 | Good default for this assignment.                  |

### Recommended risk bands



## 7. Success metrics

Accuracy alone is not enough. If late orders are less common than on-time orders, a model can look accurate while failing to catch the cases the business cares about.

| Metric                       | Minimum target | Why it matters                                        |
|------------------------------|----------------|-------------------------------------------------------|
| ROC-AUC                      | $\geq 0.75$    | Measures ranking quality across thresholds.           |
| Recall for late orders       | $\geq 0.65$    | Measures how many truly late orders are caught.       |
| Precision for predicted late | $\geq 0.55$    | Measures how many escalated orders are actually late. |
| Scored row completeness      | 120/120        | Every new order must receive a score.                 |
| Leakage control              | Required       | Model must not use post-delivery information.         |

### Strong result targets

| Metric                       | Strong target                            |
|------------------------------|------------------------------------------|
| ROC-AUC                      | $\geq 0.82$                              |
| Recall for late orders       | $\geq 0.72$                              |
| Precision for predicted late | $\geq 0.60$                              |
| Business output              | Clear risk bands and recommended actions |

### Interpretation examples

- High recall with very low precision means the model catches many late orders but creates too many false alarms.
- High precision with very low recall means escalations are usually correct but the model misses too many late orders.
- ROC-AUC helps you compare models before choosing an operational threshold.

## 8. Final Excel output specification

Your final deliverable should be a new Excel file containing one row per order in `Scoring_Orders`. The file should be easy for an operations team to filter, sort, and act on.

| Column             | Type      | Required logic                                                       |
|--------------------|-----------|----------------------------------------------------------------------|
| order_id           | Text      | Copied from <code>Scoring_Orders</code> .                            |
| late_probability   | Decimal   | Model probability for class 1. Recommended range: 0.000 to 1.000.    |
| predicted_is_late  | Integer   | 1 if late_probability is above chosen threshold, otherwise 0.        |
| risk_band          | Text      | Low, Medium, or High based on late_probability.                      |
| recommended_action | Text      | Business action based on risk band and top risk factor.              |
| top_risk_factor    | Text      | Most important observed risk driver or clear rule-based explanation. |
| model_used         | Text      | Model name, for example <code>RandomForestClassifier</code> .        |
| scored_at          | Date/time | Timestamp when scoring was produced.                                 |

### Recommended action mapping

| Risk band | Condition                       | Recommended action                                                    |
|-----------|---------------------------------|-----------------------------------------------------------------------|
| Low       | late_probability < 0.30         | No action required.                                                   |
| Medium    | 0.30 <= late_probability < 0.60 | Monitor order and prepare escalation if risk increases.               |
| High      | late_probability >= 0.60        | Prioritize driver assignment and check restaurant preparation status. |

### Example output rows

| order_id  | late_probability | predicted_is_late | risk_band | recommended_action           |
|-----------|------------------|-------------------|-----------|------------------------------|
| ORD-01001 | 0.82             | 1                 | High      | Prioritize driver assignment |
| ORD-01002 | 0.47             | 0                 | Medium    | Monitor order                |
| ORD-01003 | 0.18             | 0                 | Low       | No action required           |

## 9. Level 2 data scientist rubric

This rubric shows what the project measures. A Level 2 data scientist should be able to move from a business problem to a working, validated, and usable model output without needing a fully specified notebook.

| Skill            | Basic                  | Level 2 expected                                     | Strong                                          |
|------------------|------------------------|------------------------------------------------------|-------------------------------------------------|
| Problem framing  | Understands target     | Explains business use and prediction timing          | Connects threshold to operational cost          |
| Data cleaning    | Handles missing values | Uses consistent preprocessing pipeline               | Documents assumptions clearly                   |
| Feature handling | Uses most columns      | Removes IDs/leakage and encodes categoricals         | Creates safe useful features                    |
| Modeling         | Trains one model       | Compares baseline and stronger model                 | Justifies final model selection                 |
| Evaluation       | Reports accuracy       | Reports ROC-AUC, recall, precision, confusion matrix | Tunes threshold with business logic             |
| Delivery         | Exports predictions    | Creates clean Excel with required columns            | Adds business-friendly actions and explanations |

### What makes this Level 2 instead of Level 1

- You are expected to notice leakage without being told multiple times.
- You are expected to evaluate using metrics aligned with the business problem.
- You are expected to return a production-like Excel output, not just a notebook result.
- You are expected to explain the model output in terms a non-technical stakeholder can use.

## 10. Submission checklist

Before submitting the final Excel file, verify each item below.

| Check           | Pass criteria                                                                |
|-----------------|------------------------------------------------------------------------------|
| Rows            | Final output has exactly 120 scored rows.                                    |
| Columns         | All required output columns are present and named correctly.                 |
| Probability     | late_probability is numeric and between 0 and 1.                             |
| Prediction      | predicted_is_late is 0 or 1 only.                                            |
| Risk band       | risk_band contains only Low, Medium, High.                                   |
| Actions         | recommended_action is filled for every row.                                  |
| Leakage         | actual_delivery_time_min and target-derived fields are not used as features. |
| Metrics         | Validation ROC-AUC, recall, precision, and confusion matrix are recorded.    |
| Reproducibility | The same preprocessing is applied to validation and scoring data.            |

### Suggested short explanation to include with submission

**My final model was [model name]. I removed leakage columns, used [preprocessing method], and evaluated the model on a stratified validation split. The validation ROC-AUC was [value], late-order recall was [value], and precision was [value]. I used a threshold of [value] because [business reason]. I scored all 120 new orders and exported the required Excel output.**

# 11. Starter implementation guide

The following structure is a guide, not a mandatory solution. Adapt it to your coding style. The main point is to keep training, validation, and scoring transformations consistent.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder
from sklearn.impute import SimpleImputer
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import roc_auc_score, precision_score, recall_score, confusion_matrix

file_path = "level2_food_delivery_late_prediction_project.xlsx"
train_df = pd.read_excel(file_path, sheet_name="Train_Orders")
score_df = pd.read_excel(file_path, sheet_name="Scoring_Orders")

TARGET = "is_late"
ID_COL = "order_id"
LEAKAGE_COLS = ["actual_delivery_time_min"]

feature_cols = [c for c in train_df.columns if c not in [TARGET, ID_COL] + LEAKAGE_COLS]
X = train_df[feature_cols]
y = train_df[TARGET]

num_cols = X.select_dtypes(include="number").columns.tolist()
cat_cols = X.select_dtypes(exclude="number").columns.tolist()

preprocess = ColumnTransformer([
    ("num", SimpleImputer(strategy="median"), num_cols),
    ("cat", Pipeline([
        ("imputer", SimpleImputer(strategy="most_frequent")),
        ("onehot", OneHotEncoder(handle_unknown="ignore"))
    ]), cat_cols)
])

model = Pipeline([
    ("prep", preprocess),
    ("clf", RandomForestClassifier(n_estimators=250, random_state=42, class_weight="balanced"))
])

X_train, X_val, y_train, y_val = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42
)
model.fit(X_train, y_train)

val_prob = model.predict_proba(X_val)[: , 1]
threshold = 0.50
val_pred = (val_prob >= threshold).astype(int)

print("ROC-AUC:", roc_auc_score(y_val, val_prob))
print("Recall:", recall_score(y_val, val_pred))
```

## Starter implementation guide continued

```
print("Precision:", precision_score(y_val, val_pred))
print("Confusion matrix:\n", confusion_matrix(y_val, val_pred))

score_prob = model.predict_proba(score_df[feature_cols])[ :, 1]
output = pd.DataFrame({
    "order_id": score_df[ID_COL],
    "late_probability": score_prob,
})
output["predicted_is_late"] = (output["late_probability"] >= threshold).astype(int)
output["risk_band"] = pd.cut(
    output["late_probability"],
    bins=[-0.01, 0.30, 0.60, 1.01],
    labels=["Low", "Medium", "High"]
)
output["recommended_action"] = output["risk_band"].map({
    "Low": "No action required",
    "Medium": "Monitor order",
    "High": "Prioritize driver assignment"
})
output["top_risk_factor"] = "See model explanation"
output["model_used"] = "RandomForestClassifier"
output["scored_at"] = pd.Timestamp.now()

output.to_excel("late_delivery_predictions.xlsx", index=False)
```

## Important improvement idea

Instead of filling `top_risk_factor` with a fixed message, you can create simple rule-based explanations from the row values. For example: High traffic, Long distance, Peak hour, or High prep estimate. This is acceptable for a Level 2 project if you explain the rule.

## 12. Optional extensions

Complete these only if you finish the main project early. They are not required for Level 2 success, but they can make your submission stronger.

| Extension            | What to add                                               | Why it helps                                 |
|----------------------|-----------------------------------------------------------|----------------------------------------------|
| Threshold comparison | Test thresholds such as 0.40, 0.50, 0.60                  | Shows you understand operational trade-offs. |
| Feature importance   | Export top 10 model drivers                               | Improves explainability.                     |
| Risk dashboard       | Add a second sheet with counts by risk_band               | Makes the output easier for business users.  |
| Error analysis       | Review false positives and false negatives                | Shows mature validation thinking.            |
| Calibration check    | Compare predicted probability bands with actual late rate | Tests whether probabilities are trustworthy. |

### Final expected result

**A successful submission is a clean Excel file that operations can use immediately: every new order scored, late probability included, risk band assigned, and action recommended. Your model should also have honest validation metrics and no leakage.**

### Project files

| File                                              | Purpose                                                                        |
|---------------------------------------------------|--------------------------------------------------------------------------------|
| level2_food_delivery_late_prediction_project.xlsx | Starter workbook with train data, scoring data, output template, and guidance. |
| level2_food_delivery_project_guide.pdf            | This complete project guide.                                                   |
| late_delivery_predictions_.xlsx                   | Your final scored output file to return.                                       |